

Zhihao Chen

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EDUCATION

Nanjing University of Posts and Telecommunications (NJUPT)

Sep 2024 - Jun 2028

B.S. in Computer Science | CET-6

EXPERIENCE

Institute of Industrial Artificial Intelligence, CAS (IIAI) - Research Intern

2026 - Present

- Contribute to embodied-AI research on social navigation and robotic manipulation, building reproducible PyBullet simulation, benchmarking, and evaluation pipelines.
- Design controlled studies for navigation safety and manipulation recovery using seeded scenarios, perturbations, and quantitative evaluation across repeatable conditions.

SELECTED PROJECTS

AgentLab - LLM Agent Evaluation & Observability

- Built LLM-agent evaluation infrastructure with isolated workspaces, deterministic verification, structured traces, repeated trials, experiment comparison, and persisted results.
- Added approval-aware suspend/resume and optional LLM-as-a-Judge while keeping deterministic verification authoritative.

MCP ToolHub - Secure MCP Execution Gateway

- Built a local stdio MCP execution gateway exposing exactly 14 tools across bounded filesystem, shell, Git, audit, and control operations.
- Implemented out-of-band human approval with immutable, expiring, single-use requests, workspace confinement, replay resistance, and structured audit traces.

Repo Doctor - AI Repository Diagnosis & Repair

- Built a local-first CLI for deterministic repository diagnostics and opt-in AI repair with constrained patch generation, verification, exact-byte rollback, and optional MCP execution.
- Validated v0.3.0 with 227 tests passed, 5 skipped and a 100/100 self-scan; added approval-aware MCP suspend/resume for protected repair operations.

SocialNav-Bench - Social Navigation Safety Benchmark

- Built a reproducible PyBullet benchmark for socially-aware robot navigation with seeded scenarios, planner comparisons, and quantitative safety/navigation metrics.
- Added a predictive local safety shield that reduced collision rate from 18% to 0% (N=3), 23% to 1% (N=5), and 43% to 7% (N=10) across 100 episodes per density.

Additional projects available at github.com/zhihaochen67

TECHNICAL SKILLS

Languages: Python, C++ (basic)

AI / Agents: PyTorch, LLM Agents, MCP, Agent Evaluation

Engineering: Linux, Git, pytest

Robotics: PyBullet, Social Navigation, Robotic Manipulation